

Contact

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Top Skills

Network Services

IP/MPLS

Microwave

Languages

English

Arabic

Adam Ashraf

RF / Design / Planning / Optimisation / Two-Way Radio / IP MPLS /
EME Compliance Analyst

Melbourne, Victoria, Australia

Summary

- RF network design, planning and optimization expert on multiple technologies.
- 10+ years' experience in mobile network drive test, performance, design, planning, and optimization.
- Intensive knowledge in LTE and 5G technologies.
- Detailed knowledge of radio propagation and path prediction for communication systems.
- Solid understanding of Australian Telecommunications EME standards (ACMA/MCF/RPS3/NATA).
- Hands-on experience with using RFNSA, RF-Map2, IXUS and Visio EME design tools.
- Hands on two-way radio and P25 technology, base station components, antennas, filters, combiners and site plumbing.
- Sound knowledge of IP networks, backhaul, redundancy and availability.

Tools:

IXUS modeler, IXUS Manager & Visio design tools
Huawei U2000, PRS, OMStar, Nastar RNO, NetAct, FMA
Ericsson MSDP tools such as OSS, AMOS, ENIQ, Business object
Drive test post processing TEMS, Actix, Mapinfo

Specialties:

- Hands on IXUS modeller/manager for site modelling and EME compliance assessment
- Very familiar with UMTS/LTE optimization principles and related parameters
- Expert in the analysis on any post processing tool for any kind of drive test (Scanner, Data, Voice short calls, Voice long calls...)

Experience

Telstra

3 years 9 months

Senior Network Engineer

July 2023 - Present (2 years 10 months)

Melbourne, Victoria, Australia

- Design, integrate and deployment of new P25 sites for the land mobile radio network.
- Identifying and resolving interference causes across the SMR/RMR network.
- Perform coverage and drive survey for P25 sites across the network.
- Provide integration support for upgrades and new P25 sites.
- Apply cost-effective design solutions for the Public Safety Network to ensure solutions meet business objectives.
- Design Microwave radio links, interference, link budgets and spectrum planning.
- Configure and test Cisco IR1101 routers for LTE backhaul.
- Design and implementation of transmission systems including MPLS rings and VLAN design.
- Experience in integrating hardware and software for Microwave radio systems.
- Understanding of Microwave links, troubleshooting and resolving degraded links.

RAN Planning & Design Engineer

August 2022 - June 2023 (11 months)

Melbourne, Victoria, Australia

- Engaged in Greenfield, Augmentation, and SMC projects for 4G/5G Design.
- Reviewing CANRAD designs using CSDT for D-RAN & C-RAN sites.
- Generating start up data for 3G/4G/5G/mm-wave to facilitate sites integration.
- Resolving constraints related to site upgrades, particularly concerning structure loading and antenna space.
- Issuing Site Acquisition & Approval Briefs (SAAB) for new sites and relocations.
- Conducting desktop studies on Atoll and mapping tools using site-survey data for identification/evaluation of candidates.
- Resolving EME issues for Greenfield, SOW, and Augmentation projects.

Optus

EME Compliance Specialist

January 2020 - July 2022 (2 years 7 months)

Melbourne, Victoria, Australia

- Perform EME and regulatory compliance assessment based on RPS3 standards and MCF guidelines.
- Quality check on suppliers' various EME compliance documentation with performing mitigation actions to ensure sites compliance.
- Review and analyze detailed design documents including design feasibility reports (FR), FC reports, As built and RLM.
- Perform EME and regulatory compliance assessment for OPTUS sites.
- Ensure site design feasibility meets RPS3 standards also EME design aligns with MCF/NATA guidelines.
- Review suppliers' submitted Form A/B and PRD Preliminary RF Drawings (PRDs) designs.
- Ensure site information are correct and accurate throughout all project stages and multiple suppliers.
- EME Mitigation actions implementation and troubleshooting to ensure site compliance and minimize site shutdown with full awareness on sites EME conditions, signage and EMEG access process.
- Provide technical support for the team and assist with resolving complex site EME design issues.
- Handle and ensure ACMA restrictions are applied and implemented to minimize interference with NBN.

Huawei Technologies

1 year 6 months

EME Compliance Engineer

January 2019 - January 2020 (1 year 1 month)

Sydney, Australia

- Manage EME compliance, including EME assessment, analysis, design, technical support and performing mitigation actions to ensure site compliance for new sites and site upgrades.
- Form-A/Form-B/ RFNSA STAD and customer document database update.
- Creates Preliminary RF Drawings (PRD) using IXUS.
- Responsible for modelling and finalizing sites' EME Guide for Site Safety reports.
- Solid experience with IXUS modeler, IXUS Manager & Visio design tools.
- RFNSA configuration, ACMA registration & Plume analysis via IXUS.
- Site implementation validation, site survey & POI reports check.
- Site RF parameter generation, site activation/deactivation, RAN performance monitor and troubleshooting for GSM/UMTS/LTE/NB-IOT networks.
- VHA/EJV site engineering design and antenna solution proposal.

Senior RF RAN Engineer

August 2018 - December 2018 (5 months)

- Performance monitoring for swapped 3G/LTE sites from NSN to Huawei for Optus network.
- Optimize network violating sites' KPIs in preparation for sites acceptance.
- Troubleshoot RF faults after swap and network expansions activities.
- LTE RF interference identification, analysis, troubleshooting and mitigation after swap activity.
- Create acceptance benchmarking reports with technical analysis.
- Check RANCAP expansion activities on sites with full KPIs monitoring and reporting to customer.
- Technically work with installation team post swap process to ensure sites H/W problems are troubleshoot and solved on the spot.
- Prepare and deliver technical presentations and liaise closely with customer.
- Communicate with Huawei's EME team to ensure sites are EME compliant after any expansion activity.
- Experience in Australia's EME, MCF and RPS3 guidelines and standards.

Vodafone

2G/3G/4G LTE Radio Network Planning & Optimization Engineer

October 2014 - March 2018 (3 years 6 months)

- Expert knowledge of 2G/3G/4G technologies
- Maintain 2G/3G/4G radio network KPI's within agreed SLA's
- Apply Network audit and define new strategy for 3G/LTE networks
- RF Optimization (physical and logical parameters)
- Help performance monitoring team and define their tasks
- Worst Cells analysis and troubleshooting
- Responsible for the introduction of new features, releases, technologies into the existing radio network in multi-layer environment with existing 3G and LTE technologies
- Involved in new strategies and plans (Multi-Carrier Strategy, U900 activation)
- Neighbor planning and optimization
- Utilization and capacity studies
- Event planning / preparations
- Area ownership

Huawei Technologies

RAN Optimization Engineer

November 2012 - September 2014 (1 year 11 months)

- Daily KPIs monitoring for 3G network and analyzing KPIs trends.
- Optimization & troubleshooting of the 3G top worst problematic cells
- Enhancing accessibility KPIs [CS, R99 & HSDPA CSSR]
- Recommend and follow up on RF capacity upgrades to ensure congestion levels are kept to a minimum
- Customer complaints
- Customer interfacing
- Organize technical workshops and training

Nokia

Radio Network Field Engineer

September 2012 - October 2012 (2 months)

- Preparation 2G, 3G and HSDPA Clusters drive test analysis reports using TEMS 10.0.5, MapInfo and ACTIX including post processing
- Detect & analyze drive test problems
- Performing 2G, 3G sites coverage and troubleshooting
- Handling optimization tasks
- Site audit (azimuth, mechanical tilt and electrical tilt)
- Troubleshooting and handling customer complains
- Performing HSDPA tests for specific VIP sites
- Dealing with both (Normal and Concentric) Cells

Education

Modern Academy for Engineering & Technology

Bachelor's Degree, Telecommunication Engineering · (2007 - 2012)